

Roll No.....

Mid Semester Examination

Name of the Program: B.Tech.

Semester: 1

Name of the Course: Engineering Chemistry

Course Code: TCH101

Time: 90 Minutes

Maximum Marks: 50

Note:

- i Answer **all the questions** by choosing **any one of the sub questions**.
- ii Each question carries 10 marks

Q1	(10 marks)	CO1
(a)	i. Differentiate between intramolecular and intermolecular hydrogen bonding. ii. Draw completely labelled MOT diagram of O_2^+ and N_2^+ and explain their magnetic property	
OR		
(b)	Explain different types of vibrations involved in IR Spectroscopy. Write applications of IR Spectroscopy	
Q2	(10 marks)	CO1
(a)	i. Explain Band Theory with discussion on conductors, semi-conductors and insulators. Give a diagrammatic representation ii. Enumerate the metallic properties of a metal	
OR		
(b)	Using the molecular orbital theory, compare the bond order and magnetic character of NO, NO^+ , NO^- and give stability order.	
Q3	(10 marks)	CO2
(a)	i. Describe the principle of ion exchange process of boiler water treatment with appropriate diagram and chemical equations. ii. Outline 2 advantages of the ion exchange process vis-à-vis lime soda process of water treatment	
OR		
(b)	Calculate the amount of lime and soda required for the treatment of 100,000 litres of water containing the following impurities in ppm: $Mg(HCO_3)_2 = 20.8$, $CaCl_2 = 28.1$, $MgCl_2 = 8.78$, $Ca(HCO_3)_2 = 30.2$, $CaSO_4 = 35.0$, $MgSO_4 = 6.7$. The purity of lime and soda used are 70% and 85% respectively	
Q4	(10 marks)	CO2
(a)	Differentiate between i. Cold lime soda and hot lime soda ii. Temporary hardness and Permanent hardness	
OR		
(b)	A completely exhausted zeolite softener requires 120 litres of NaCl solution having 100g/l of NaCl. How many litres of water having hardness 500 ppm can be softened by the zeolite softener.	
Q5	(10 marks)	CO2
(a)	Write note on i. Scale and Sludge ii. Caustic embrittlement iii. Cation and anion exchange resin iv. Calgon conditioning	
OR		
(b)	Two water samples contain 408 mg of $CaSO_4$ per liter (solution-1) and 408 mg of $CaCl_2$ per liter (solution-2). Express their hardness in $CaCO_3$ equivalents and state which solution has a higher hardness.	

